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BERLIN SCHOOL OF BUSINESS & INNOVATION

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Leveraging Machine Learning Approaches for Stroke Risk Prediction

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Name:

Rohanta Dinkar Bhamare

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INTRODUCTION

Machine learning is the important part of AI. It works on supervise as well as unsupervised data. Not only that it also works with continuous and classification dataset. In this paper we focused on the supervise data with classification problem.

Data set is related to medical field in which consist of 12 columns and 5110 records. Features of the dataset is gender, age, hypertension, heart disease, marital status, work type, residence type, average glucose level, body mass index and smoking status. The target variable is stroke which indicates whether a patient has experienced a stroke i.e. 1 or not i.e. 0. In the further steps we are going find and handle null values and outliers. Also, if data is imbalance, then SMOTE is going to use. In the last with the train test split different machine learning model are going to train with hyperparameter tunning.

CHAPTER ONE EDA

In the EDA we take the incites of the dataset by using different libraries. First, we check shape and null values of the data set. Shape of the data set is (5510,12) and there are 201 missing values in the body mass index column. After that mean mode and other details for numerical column is checked as shown in the figure 1.1.

```
In [109]: df.describe()
```

Out[109]:

	id	age	hypertension	heart_disease	avg_glucose_level	bmi	stroke
count	5110.000000	5110.000000	5110.000000	5110.000000	5110.000000	4909.000000	5110.000000
mean	36517.829354	43.226614	0.097456	0.054012	106.147677	28.893237	0.048728
std	21161.721625	22.612647	0.296607	0.226063	45.283560	7.854067	0.215320
min	67.000000	0.080000	0.000000	0.000000	55.120000	10.300000	0.000000
25%	17741.250000	25.000000	0.000000	0.000000	77.245000	23.500000	0.000000
50%	36932.000000	45.000000	0.000000	0.000000	91.885000	28.100000	0.000000
75%	54682.000000	61.000000	0.000000	0.000000	114.090000	33.100000	0.000000
max	72940.000000	82.000000	1.000000	1.000000	271.740000	97.600000	1.000000

Figure 1.1 details of each numeric column

To find distribution and outliers of the numeric column in the further step histogram and boxplot is applied on those columns.

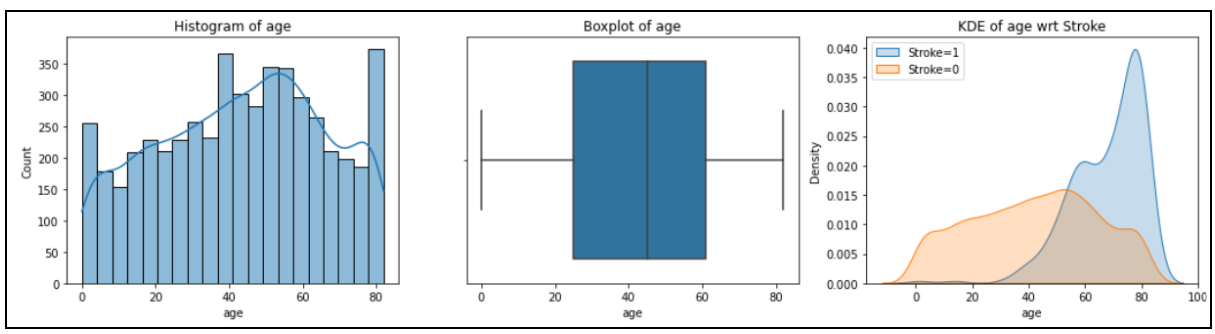

Figure 1.2 Age

From the figure 1.2 it can be concluding the possibility of having stoke is more in between if age of the patient is more than 40.

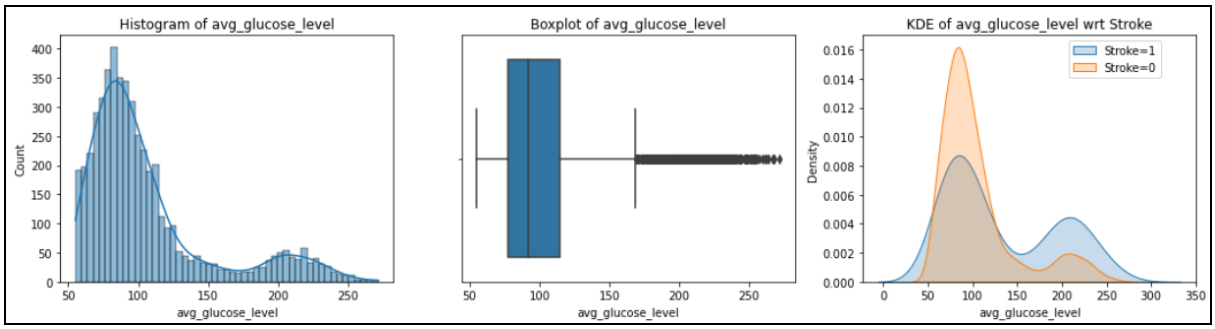

Figure 1.3 average glucose level

Image 1.3 show that this column is rightly skewed. There are outliers in the upper bond. For handing the rightly skewed data mostly logarithmic transformation or winsorization is used.

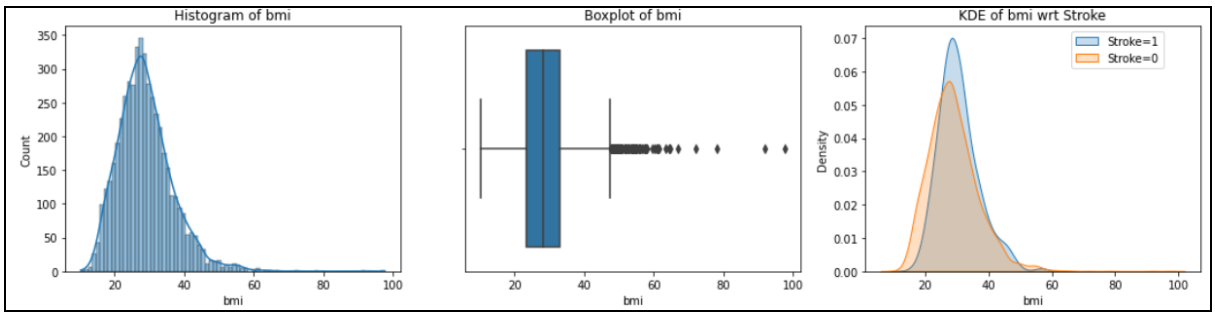

Figure 1.4 Body Mass Index

In the body mass index column, there are outliers on the upper bond it means the data is rightly skewed. As discuss earlier logarithmic transformation or winsorization is best for rightly skewed data (West, 2021).

To find the correlation of the target column with all numeric column correlation matrix is plotted.

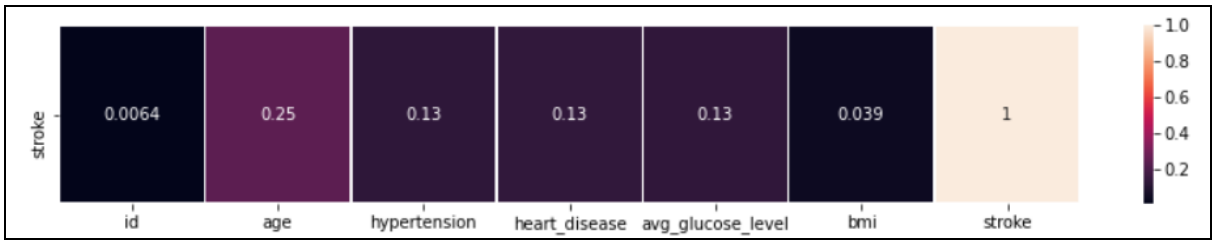

Figure 1.5 Correlation matrix

Correlation matrix shows that age is partially correlate with the target column and other are not weakly correlated as compare to the age.

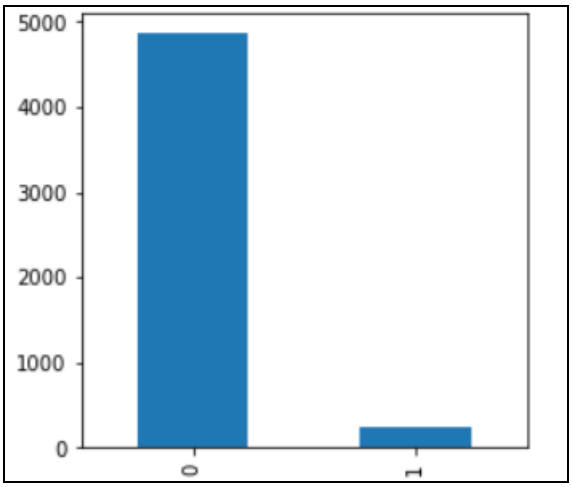

Figure 1.6 Value counts of target column

In the final the final step of EDA we checked for value counts of the target column to check weather dataset is unbalance or not. Through the figure 1.6 it can be concluded that the dataset is imbalanced toward 0. Model trained with this data set can be bias toward 0. So, after the model training SMOTE technology should be used to balance the dataset (Chawla et al., 2002).

CHAPTER TWO DATA PREPARATION

In this chapter we have to prepare data for model training. Data preparation includes handing of missing value and outliers. Also, feature engineering in that new features are created by using encoding. After that data splitting is done for training and testing. SMOTE technology is applied on the train dataset. In the last standardization is done on both train and test dataset to make the standard deviation same of all columns.

In the EDA we checked that there are some missing values in BMI column. After calculation its seems that total percentage of missing values are 3.93% of the whole dataset. The size of the dataset is small so we decided to replace null values with mean of that column.

There are outliers in two columns in the given dataset. Data was rightly skewed so winsorization technique is applied. Outliers from above 87th percentile are replaced with 87th percentile value and for lower limit below 5th percentile value replace with 5th percentile value.

In the next step feature engineering is done. In the feature engineering one hot encoding is done on object column so that all columns are converted into numeric columns.

After encoding now data is cleaned and ready for train test split. Size of dataset is small due to that train test split ratio is taken as 85/15. It means 85% for training and 15% for testing.

After train test split SMOTE is applied because data set is imbalanced. There are chances of data leakage so the SMOTE only applied for training dataset.

```
Befor SMOTE :
 0    4861
1     249
Name: stroke, dtype: int64
After SMOTE :
 0    4141
1    4141
Name: stroke, dtype: int64
```

Figure 2.1 SMOTE

After balancing the data set standardization is done. In the standardization all the column's standard deviation are convert to one. It means all the columns are on the one scale. So, at the time model training all the features are contribute the same or calculate on the same scale (Jain et al., 2005).

CHAPTER THREE MODEL TRAINNING

By using clean data different classification machine learning models are trained. Those models are Logistic Regression, Support Vector Machine, Decision Tree, Random Forest and K-Nearest Neighbors. In the training process default hyperparameters are used and stored in the form of dictionary for the further reference (Kotsiantis et al., 2007).

```
Logistic Regression:
{'C': 1.0, 'class_weight': None, 'dual': False, 'fit_intercept': True, 'intercept_scaling': 1, 'l1_ratio': None, 'max_iter': 10
00, 'multi_class': 'auto', 'n_jobs': None, 'penalty': 'l2', 'random_state': 42, 'solver': 'lbfgs', 'tol': 0.0001, 'verbose': 0,
'warm_start': False}


SVM:
{'C': 1.0, 'break_ties': False, 'cache_size': 200, 'class_weight': None, 'coef0': 0.0, 'decision_function_shape': 'ovr', 'degre
e': 3, 'gamma': 'scale', 'kernel': 'rbf', 'max_iter': -1, 'probability': True, 'random_state': 42, 'shrinking': True, 'tol': 0.
001, 'verbose': False}


Decision Tree:
{'ccp_alpha': 0.0, 'class_weight': None, 'criterion': 'gini', 'max_depth': None, 'max_features': None, 'max_leaf_nodes': None,
'min_impurity_decrease': 0.0, 'min_samples_leaf': 1, 'min_samples_split': 2, 'min_weight_fraction_leaf': 0.0, 'random_state': 4
2, 'splitter': 'best'}


Random Forest:
{'bootstrap': True, 'ccp_alpha': 0.0, 'class_weight': None, 'criterion': 'gini', 'max_depth': None, 'max_features': 'sqrt', 'ma
x_leaf_nodes': None, 'max_samples': None, 'min_impurity_decrease': 0.0, 'min_samples_leaf': 1, 'min_samples_split': 2, 'min_wei
ght_fraction_leaf': 0.0, 'n_estimators': 100, 'n_jobs': None, 'oob_score': False, 'random_state': 42, 'verbose': 0, 'warm_star
t': False}


KNN:
{'algorithm': 'auto', 'leaf_size': 30, 'metric': 'minkowski', 'metric_params': None, 'n_jobs': None, 'n_neighbors': 5, 'p': 2,
'weights': 'uniform'}
```

Figure 3.1 Default Hyperparameters for all models

CHAPTER FOUR MODEL EVALUATION

In the model evaluation all models are evaluate on the basis of accuracy score, precision, recall, F1 score and AUC. Training testing accuracy and AUC should be near to one this is the definition of perfect model. But if training accuracy is more and testing accuracy is less it means model stuck in to overfitting. So, after model training hyper parameter tunning must done to get perfect accuracy of the model. After tunning for understanding the contribution of each feature SHAP technique is used (Lundberg and Lee, 2017).

For the given dataset confusion matrix are plotted for each model with training and testing accuracy.

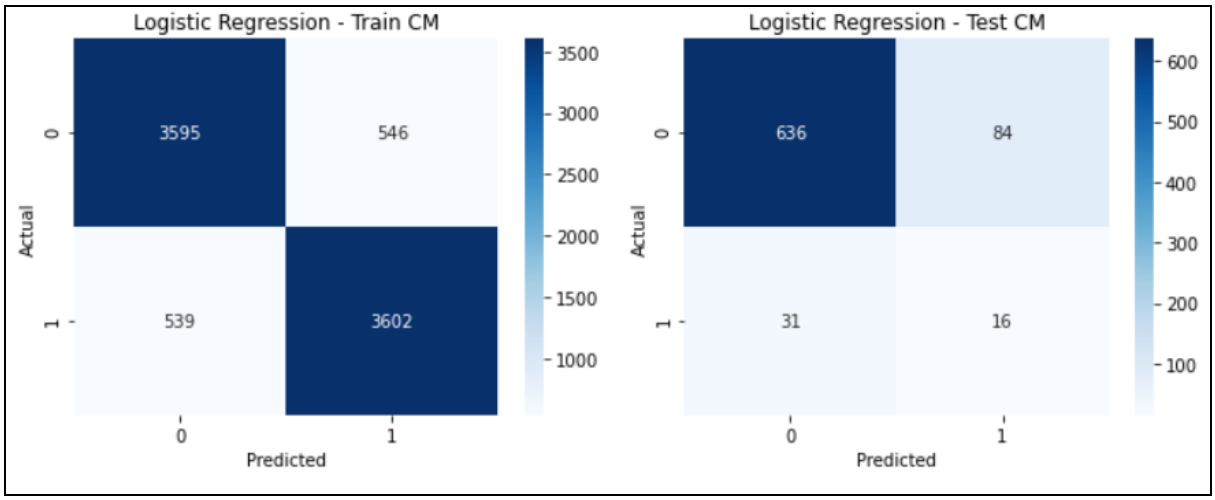

Figure 3.2 confusion matrix of Logistic Regression

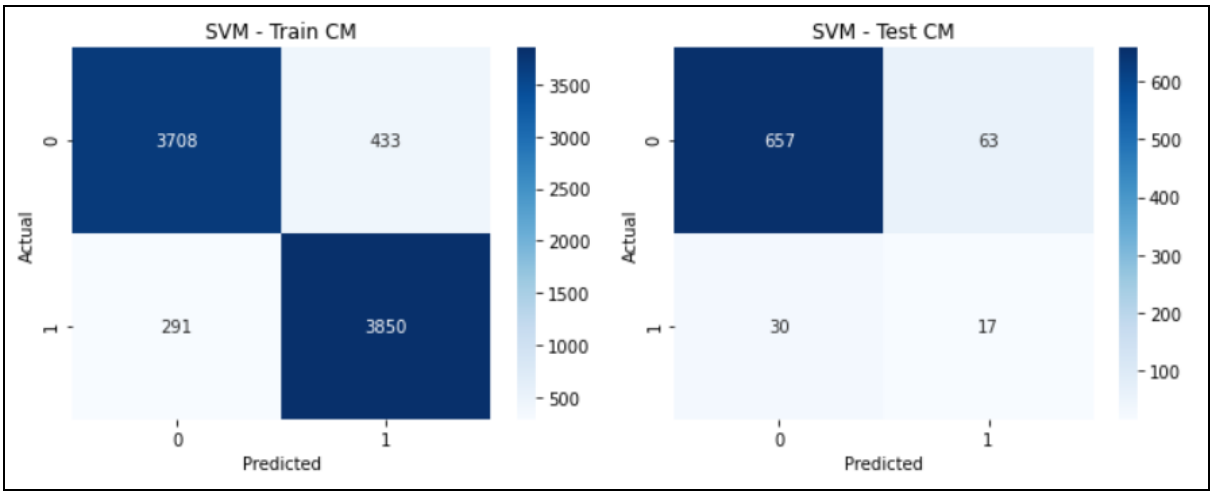

Figure 3.3 confusion matrix of SVM

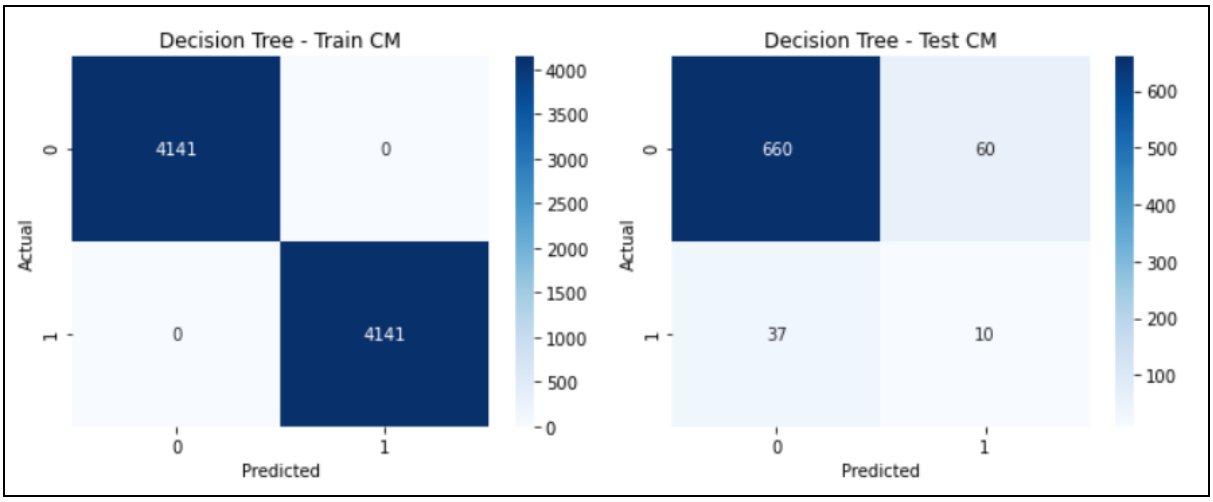

Figure 3.4 confusion matrix of Decision Tree

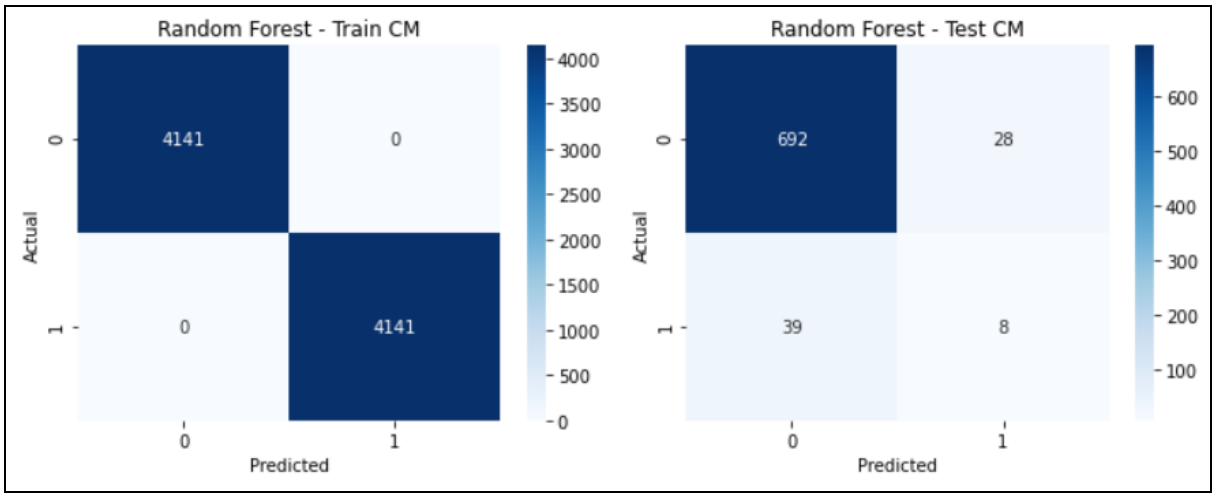

Figure 3.5 confusion matrix of Random Forest

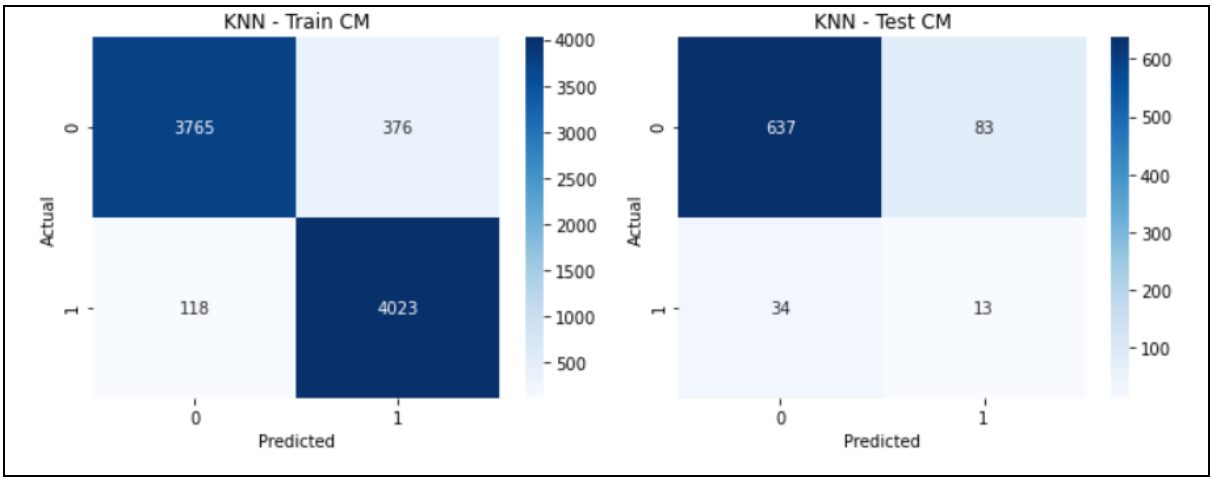

Figure 3.6 confusion matrix of KNN

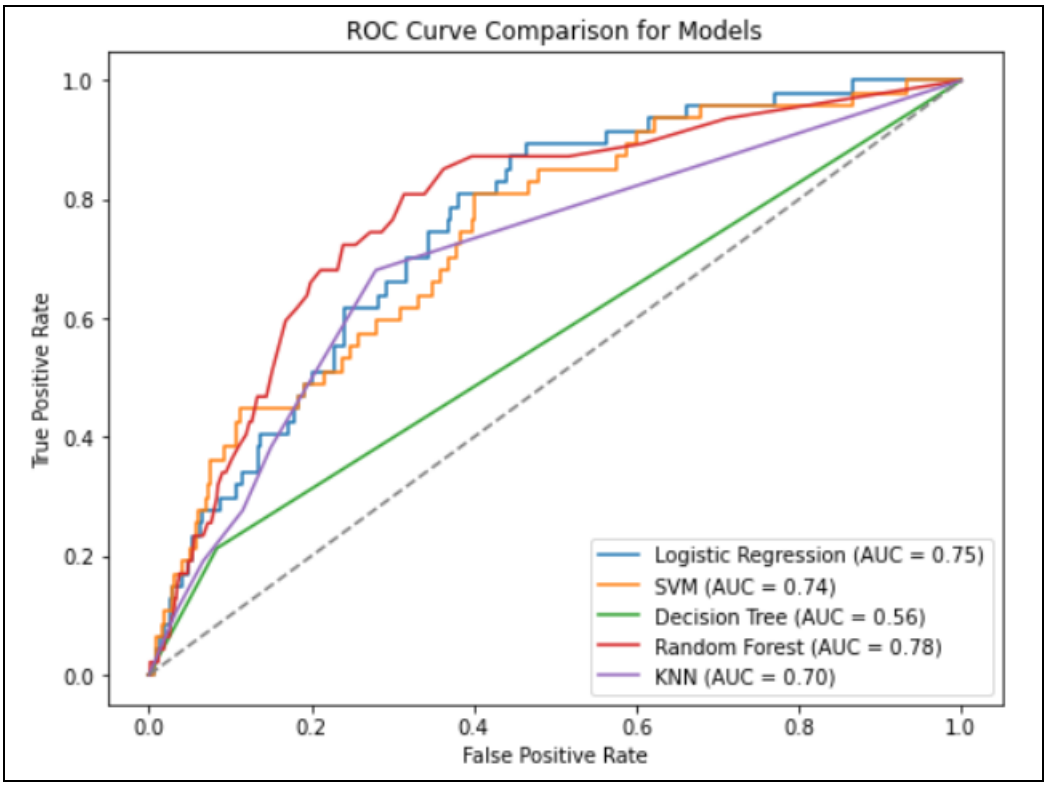

Figure 3.7 ROC AUC Curve for all model

Model	**Train Acc**	**Test Acc**	**Precision**	**Recall**	**F1**	**AUC**
Logistic Regression	0.869	0.850	0.16	0.34	0.218	0.754
SVM	0.913	0.879	0.213	0.36	0.268	0.7397
Decision Tree	1.000	0.874	0.143	0.21	0.171	0.565
Random Forest	1.000	0.913	0.222	0.17	0.193	0.781
KNN	0.940	0.847	0.135	0.28	0.182	0.704

Table 3.1 Comparison Table of accuracy of all models

For deciding best model lots of criteria's need to consider. First and very important criteria is accuracy. From the given model testing accuracy of random forest is maximum 0.913 followed by SVM with 0.879. But in the medical field recall is important so recall of SVM and logistic is way better than random forest. Also, area under the curve (AUC) is good for random forest, SVM and logistic regression. So, with the hyperparameter tunning these three models are may be perform well. On the other hand, Decision tree shows overfitting and also AUC is 0.565 which is not so good. And KNN perform well as compare to decision tree but other three are good in all parameters so we decided to move with SVM, Random Forest and Logistic Regression for hyperparameter tunning.

After the hyperparameter tunning random forest perform well.

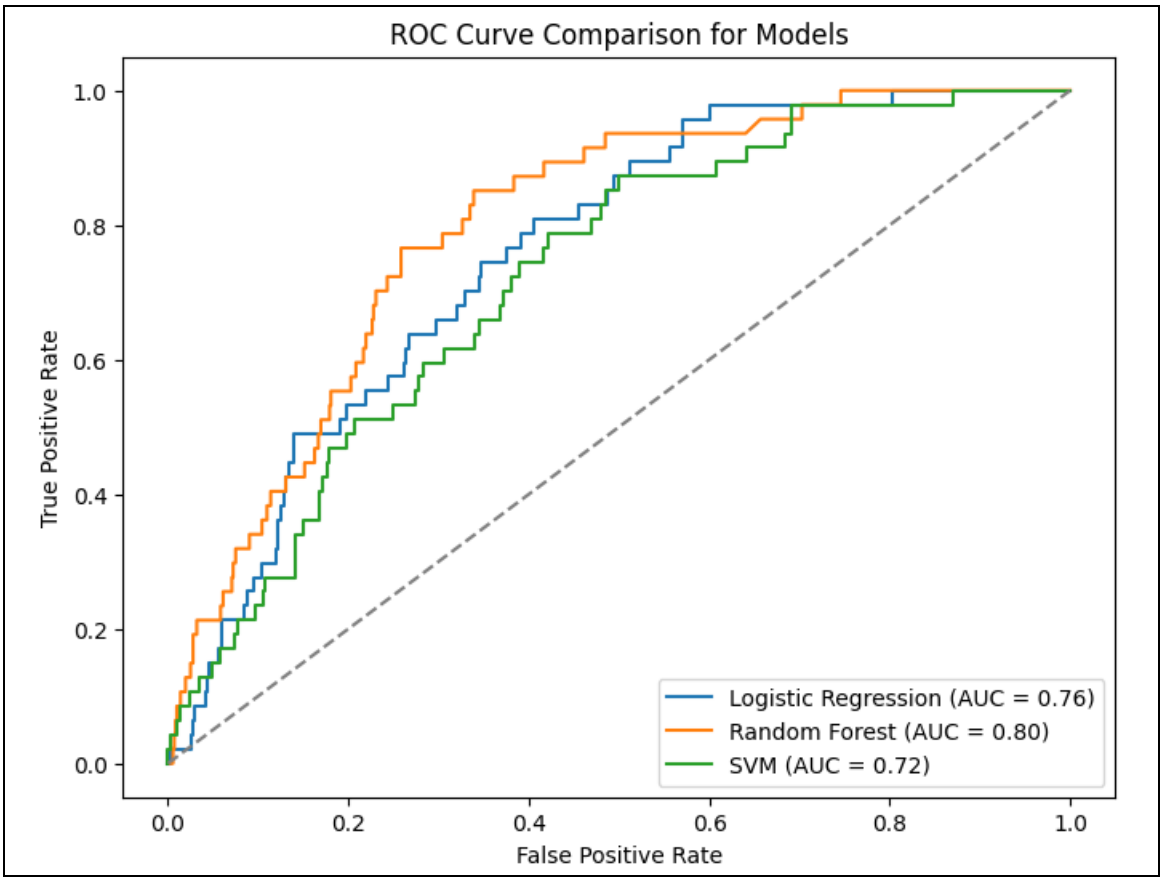

Figure 3.8 ROC AUC Curve for tunned model

Model	**Train Accuracy**	**Test Accuracy**	**Precision**	**Recall**	**F1-Score**	**AUC**
Logistic Regression	0.873	0.838	0.176	0.447	0.253	0.757
Random Forest	0.995	0.923	0.286	0.170	0.213	0.798
SVM	0.959	0.847	0.135	0.277	0.182	0.723

Table 3.2 Comparison Table of accuracy of tunned models

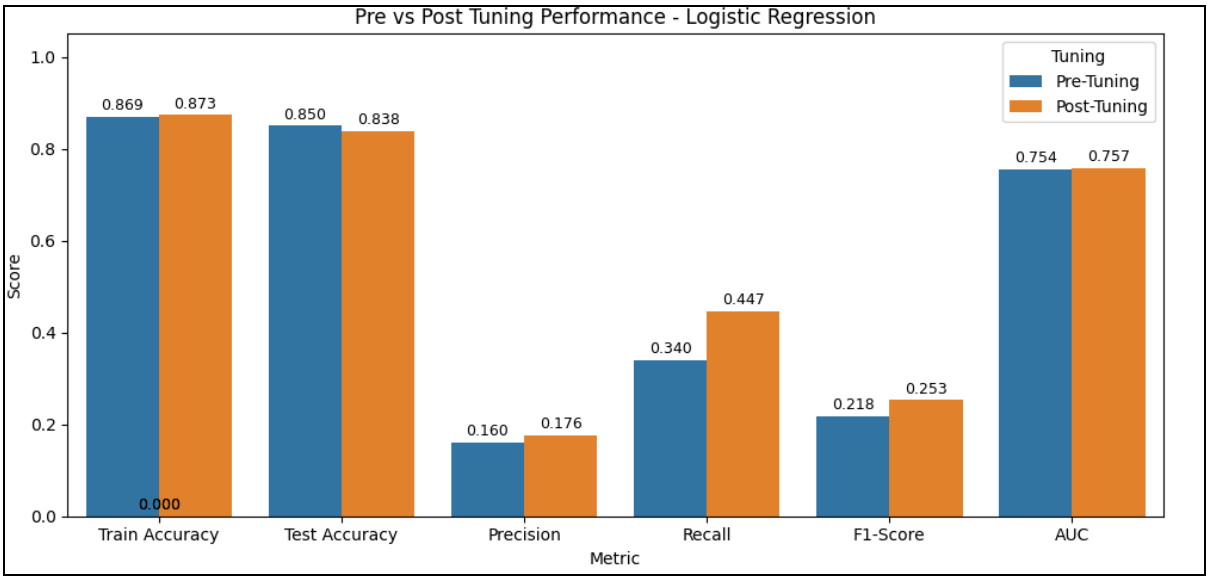

Figure 3.9. Pre vs Post Performance of Logistic Regression

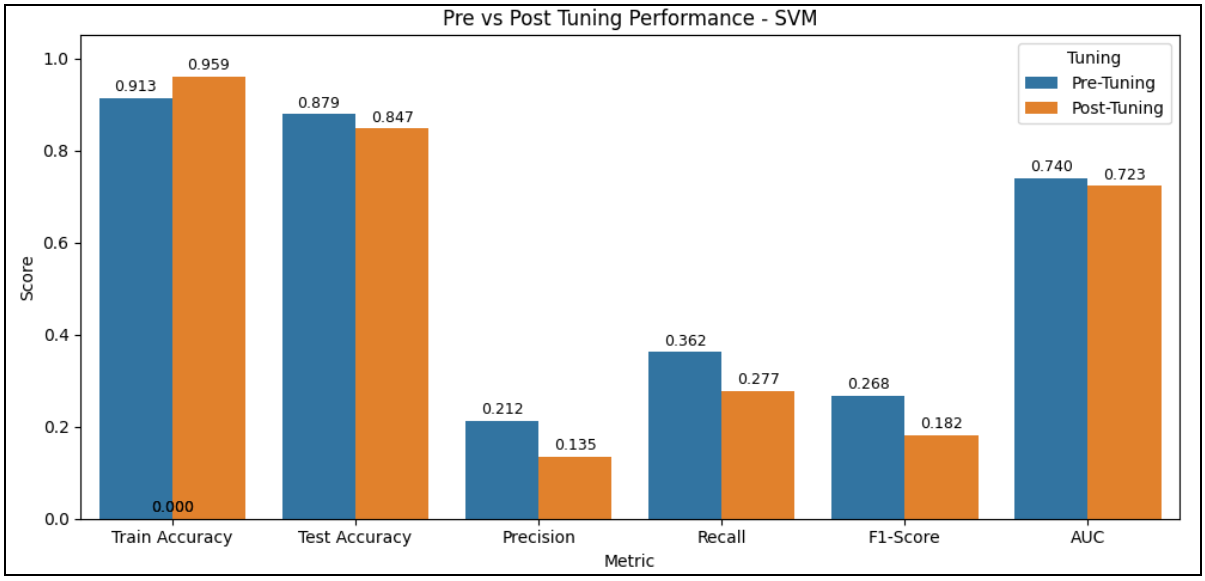

Figure 3.10. Pre vs Post Performance of SVM

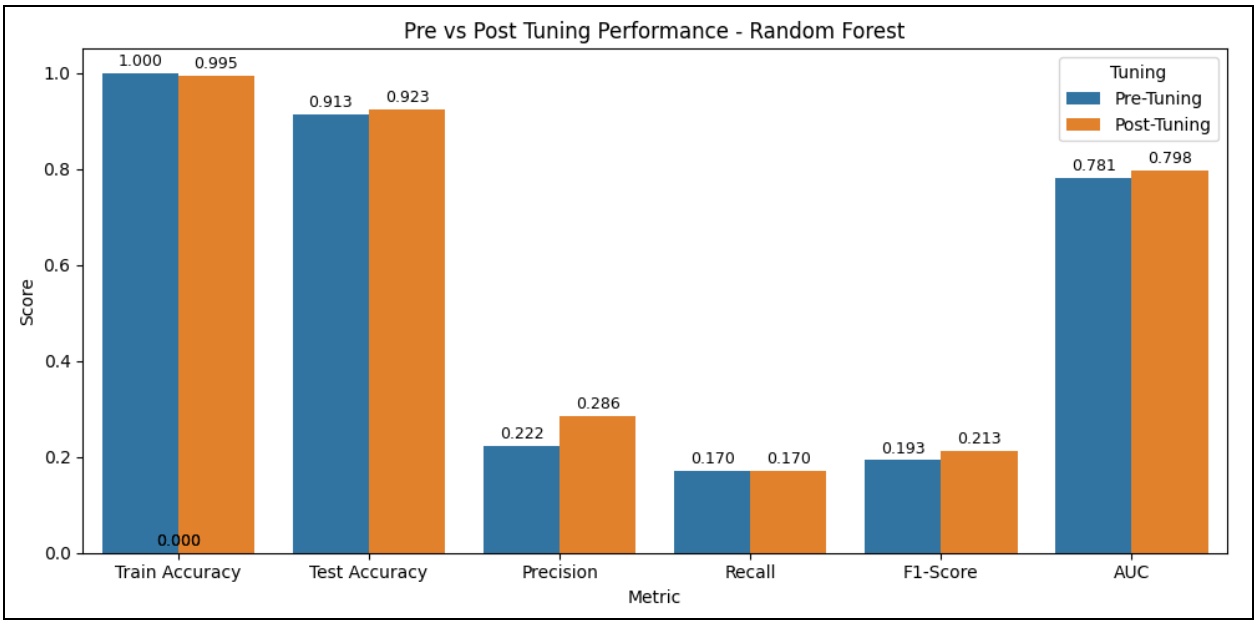

Figure 3.11. Pre vs Post Performance of Random Forest

Above plots shows Random Forest perform well as compare to other classification models. Tranning accuracy of RF is reduced and testing accuracy is increased it means we are able to overcome problem of overfitting. Also, AUC increased after tunning from 0.78 to 0.79.

CONCLUSION AND FUTURE WORK

From the above study it can be concluded that random forest performs well as compared to other model after tunning. It gives training accuracy of 0.995 and testing accuracy of 0. 923.Recall is slightly low 0.170 but AUC is good 0.798.

Size of the data set was small because of that lots of limitations was there. Also, data set was imbalance still by using SMOTE technology we are trying to balance the data set and train model as good as possible.

In the future work artificial neural network can use instead of machine learning model. But at that time size of dataset also needs to increase. Because ANN perform well only on large number of datasets.

For the reference link of Google Collab file is added below:

https://colab.research.google.com/drive/1Jtd6jc2k_t1g6d34HVQSIMec3lBfh4HM?usp=sharing

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